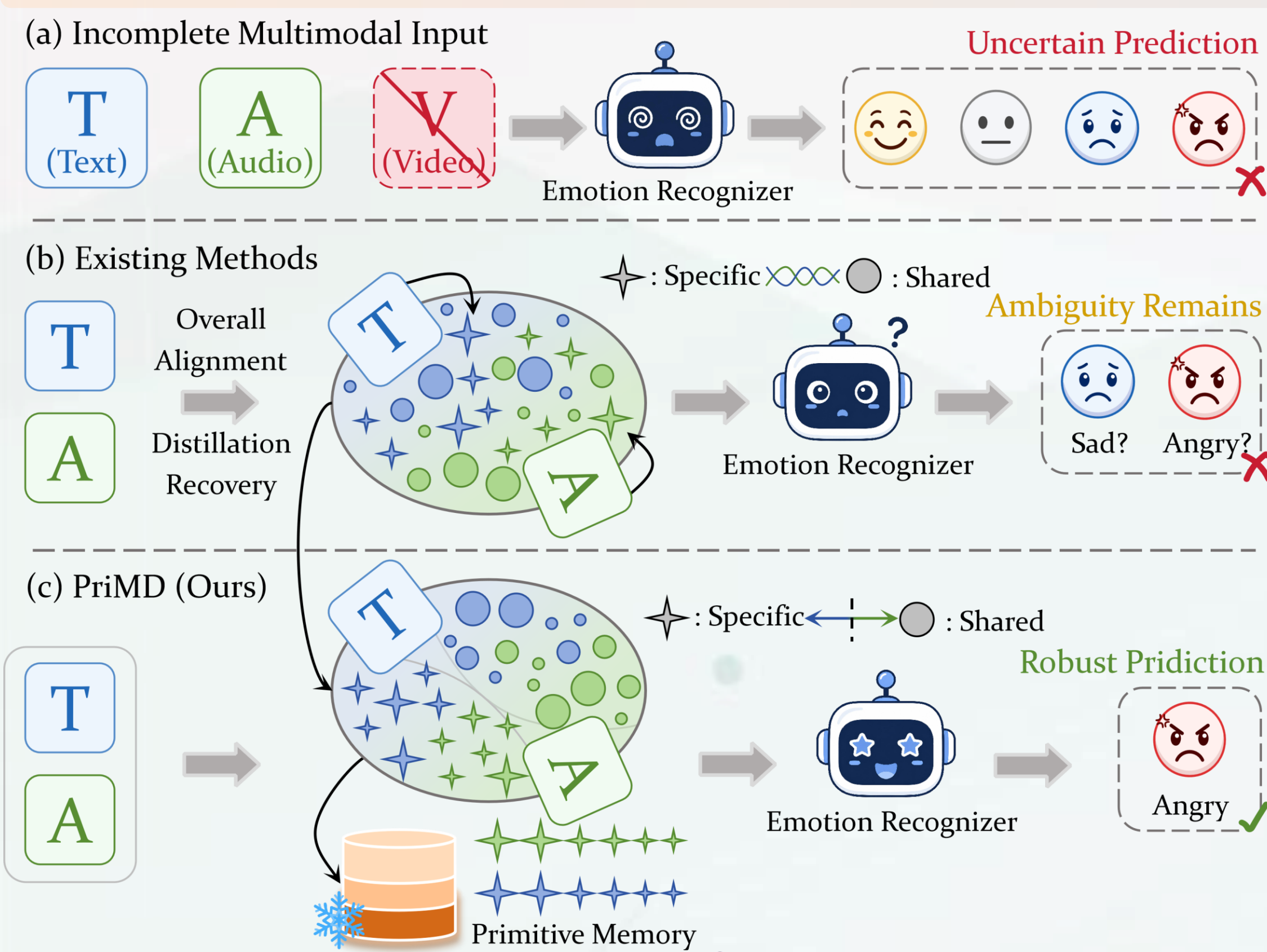


Modality Disentangled Learning for Incomplete Multimodal Emotion Recognition: A *Primitive Memory Distillation* Perspective

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Introduction & Motivation

Want to Talk More?



Challenges: Modality-specific information in missing modalities is difficult to infer reliably.

Limitations of Current Methods:

Existing methods typically handle missing modalities through holistic generation, alignment, or distillation, without distinguishing inferable shared semantics from uncertain modality-specific information.

Highlights: PriMD revisits incomplete multimodal emotion recognition from an intra-modal perspective and formulates it as a unified process of shared-semantic estimation and constrained compensation of modality-specific information.

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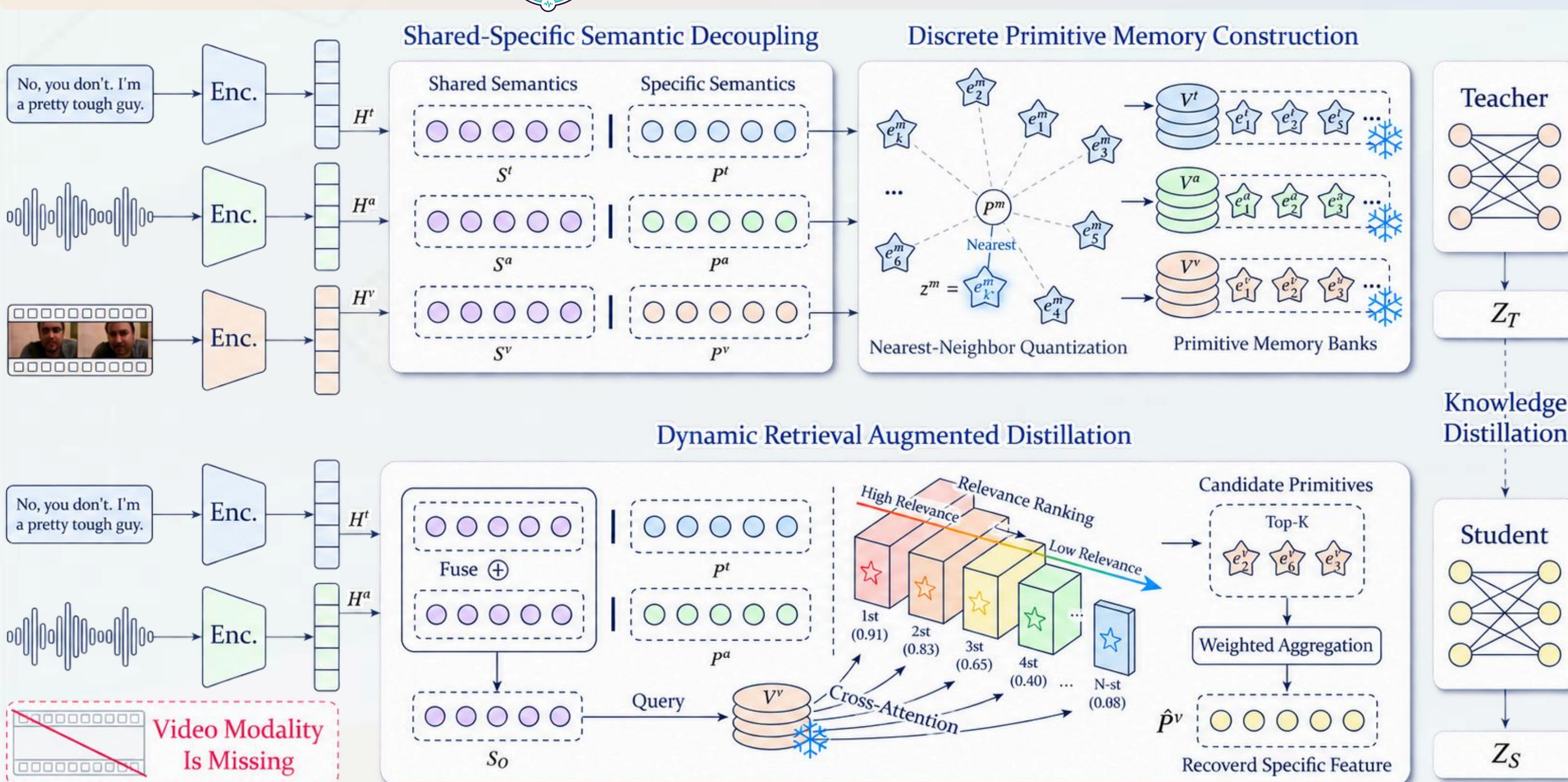
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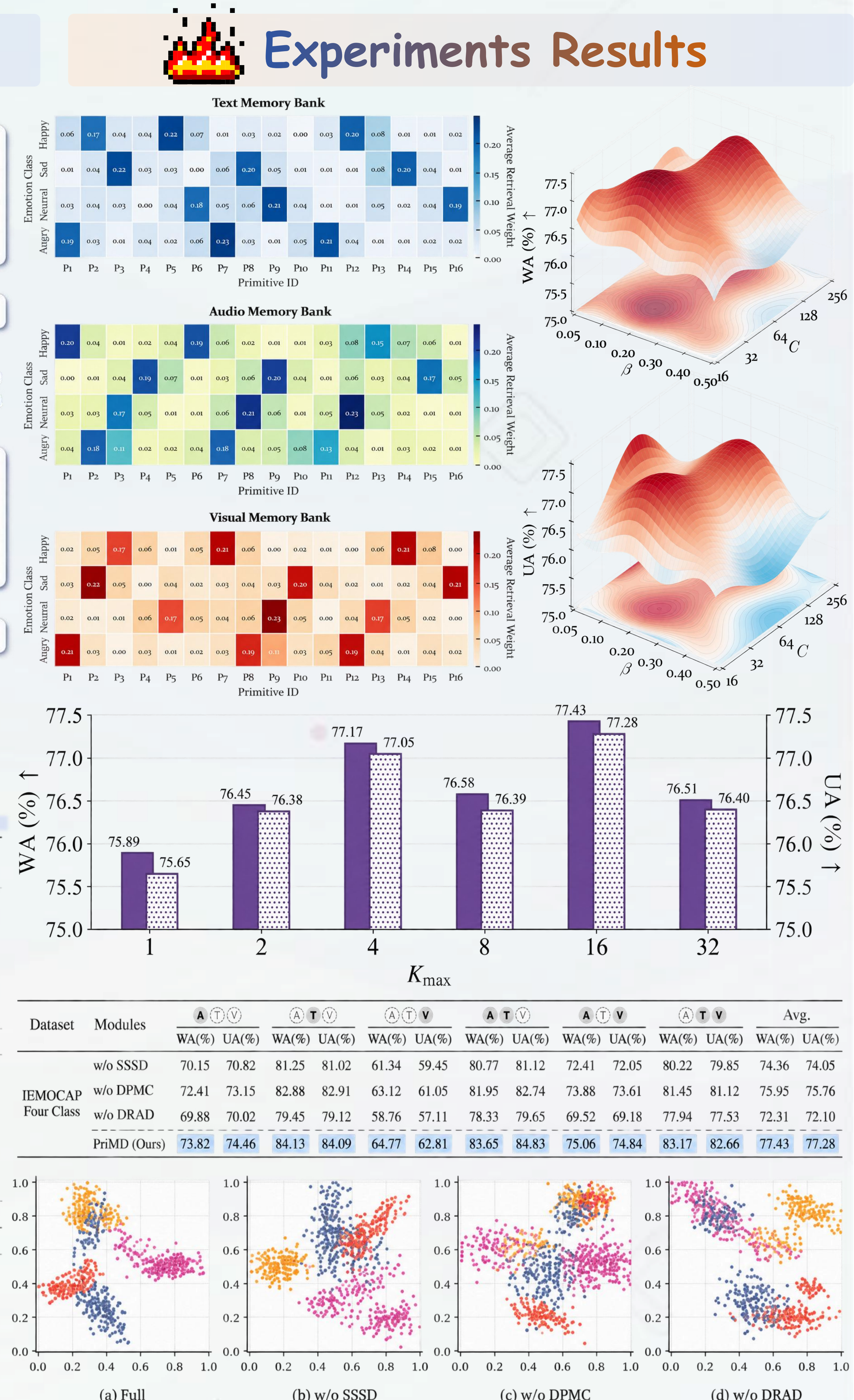
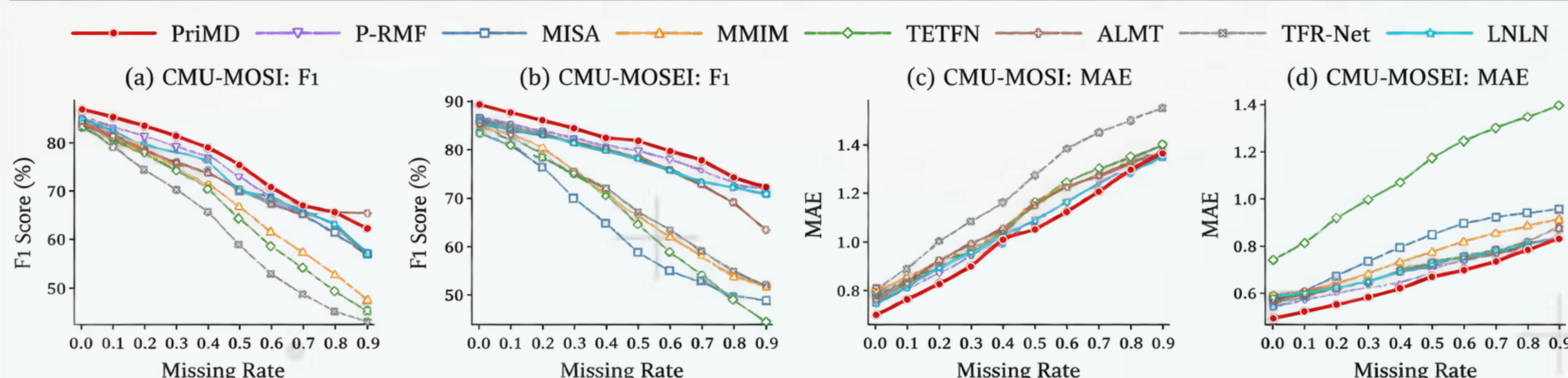
Code

Method: PriMD



We design SSSD to disentangle shared and modality-specific features, and DPMC/DRAD to discretize the modality-specific space into semantic primitives and retrieve them for compensation under missing modalities.

Dataset	Method	A T V		A T V		A T V		A T V		A T V		A T V		Avg.	
		WA(%)	UA(%)	WA(%)	UA(%)	WA(%)	UA(%)	WA(%)	UA(%)	WA(%)	UA(%)	WA(%)	UA(%)	WA(%)	UA(%)
IEMOCAP Four Class	CPMNet (Zhang et al., 2022)	46.85	51.72	45.63	45.32	44.95	44.49	34.81	36.23	48.67	49.33	45.62	46.57	44.42	45.61
	MMIN (Zhao et al., 2021)	56.58	59.00	66.57	68.02	52.52	50.60	72.94	71.14	63.99	63.43	71.67	68.61	64.05	63.47
	GCNet (Lian et al., 2023)	65.58	68.76	72.33	70.42	57.96	52.54	77.02	76.87	67.40	65.64	75.63	73.62	69.32	67.98
	CIF-MMIN (Liu et al., 2024)	57.53	60.06	67.22	68.99	53.46	51.56	74.19	72.59	64.99	63.53	72.40	69.91	64.97	64.44
	MoMKE (Xu et al., 2024)	69.53	70.21	77.30	77.66	56.80	52.03	79.03	79.88	68.57	66.22	75.55	74.18	71.13	70.03
	HARDY-MER (Liu et al., 2025)	72.65	73.87	82.49	82.69	63.19	60.54	81.67	82.43	74.19	74.50	79.18	78.51	75.56	75.42
	PriMD (Ours)	73.82	74.46	84.13	84.09	64.77	62.81	83.65	84.83	75.06	74.84	83.17	82.66	77.43	77.28
IEMOCAP Six Class	CPMNet (Zhang et al., 2022)	29.47	29.80	32.44	34.95	26.20	24.95	33.49	33.94	26.92	25.46	31.34	30.43	29.98	29.92
	MMIN (Zhao et al., 2021)	44.08	42.96	42.17	38.55	35.74	30.65	51.95	48.31	41.92	38.15	47.49	40.63	43.89	39.88
	GCNet (Lian et al., 2023)	49.95	46.45	56.48	55.62	39.78	34.97	58.24	57.25	47.57	43.31	57.43	54.66	51.58	48.71
	CIF-MMIN (Liu et al., 2024)	44.96	43.56	43.40	39.71	36.11	31.35	52.43	49.20	42.54	39.22	48.88	44.91	44.72	41.33
	MoMKE (Xu et al., 2024)	50.51	47.38	61.09	60.19	39.07	34.51	63.18	61.94	48.65	44.08	59.92	57.55	53.74	50.94
	HARDY-MER (Liu et al., 2025)	51.58	49.14	65.89	61.95	43.02	36.49	65.18	62.98	52.91	47.45	61.66	57.86	56.71	52.65
	PriMD (Ours)	53.62	51.43	66.21	62.17	45.35	39.86	65.78	63.66	54.17	49.82	63.65	60.77	58.13	54.62
CMU MOSEI		ACC(%)	F1(%)	ACC(%)	F1(%)	ACC(%)	F1(%)	ACC(%)	F1(%)	ACC(%)	F1(%)	ACC(%)	F1(%)	ACC(%)	F1(%)
	CPMNet (Zhang et al., 2022)	65.71	65.18	72.87	72.44	61.23	61.73	72.65	72.24	61.56	61.99	66.29	66.84	66.72	66.74
	MMIN (Zhao et al., 2021)	58.90	59.50	82.20	82.40	59.30	60.01	83.70	83.30	63.55	61.91	81.75	81.42	71.57	71.42
	GCNet (Lian et al., 2023)	72.04	70.34	84.26	84.17	68.08	67.25	85.10	85.10	71.49	69.96	84.74	84.54	77.62	76.89
	CIF-MMIN (Liu et al., 2024)	63.87	64.60	83.53	83.04	61.96	62.66	84.01	83.47	64.68	62.08	82.50	81.94	73.43	72.97
	MoMKE (Xu et al., 2024)	72.56	71.03	86.10	86.03	64.50	63.46	86.32	86.29	72.37	72.07	86.90	86.91	78.13	77.63
	CMAD (Zhuang et al., 2025)	63.00	60.80	86.10	86.00	65.70	64.40	86.30	86.20	65.60	64.80	86.40	86.40	75.52	74.77
	HARDY-MER (Liu et al., 2025)	74.82	74.11	87.20	87.13	69.35	67.50	85.42	85.01	74.82	74.11	85.72	85.39	79.56	78.88
	PriMD (Ours)	74.61	74.25	87.36	87.24	68.96	67.36	87.26	87.69	73.85	73.61	86.81	87.27	79.81	79.57



Extensive experiments on three benchmarks show that PriMD outperforms prior methods under diverse missing-modality settings, with especially large gains when two modalities are absent. These results demonstrate that PriMD can effectively mitigate representation instability caused by holistic inference of missing modalities.